ASSIGNMENT 1

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SLOT :A

COURSE : C PROGRAMMING

QUESTION 1:

Aim:

To calculate the final cab fare based on distance travelled and membership discount.

Algorithm:

1. Start.
2. Read distance and membership status.
3. Set base fare to ₹ 100.
4. If distance is greater than 10 km, add ₹ 20 for each extra kilometer.
5. If the customer is a member, apply a 10% discount.
6. Display the final fare.
7. Stop.

Pseudocode:

```
START

READ distance, member

fare = 100

IF distance > 10
   fare = fare + (distance - 10) * 20
END IF

IF member == true
   fare = fare * 0.9
END IF
```

PRINT fare

STOP

C Program:

```
#include <stdio.h>

int main() {

    int distance, member;

    float fare = 100;



    scanf("%d", &distance);

    scanf("%d", &member);



    if (distance > 10)

        fare += (distance - 10) * 20;



    if (member == 1)

        fare *= 0.9;



    printf("Final Fare: Rs.%.0f", fare);



    return 0;

}
```

Result

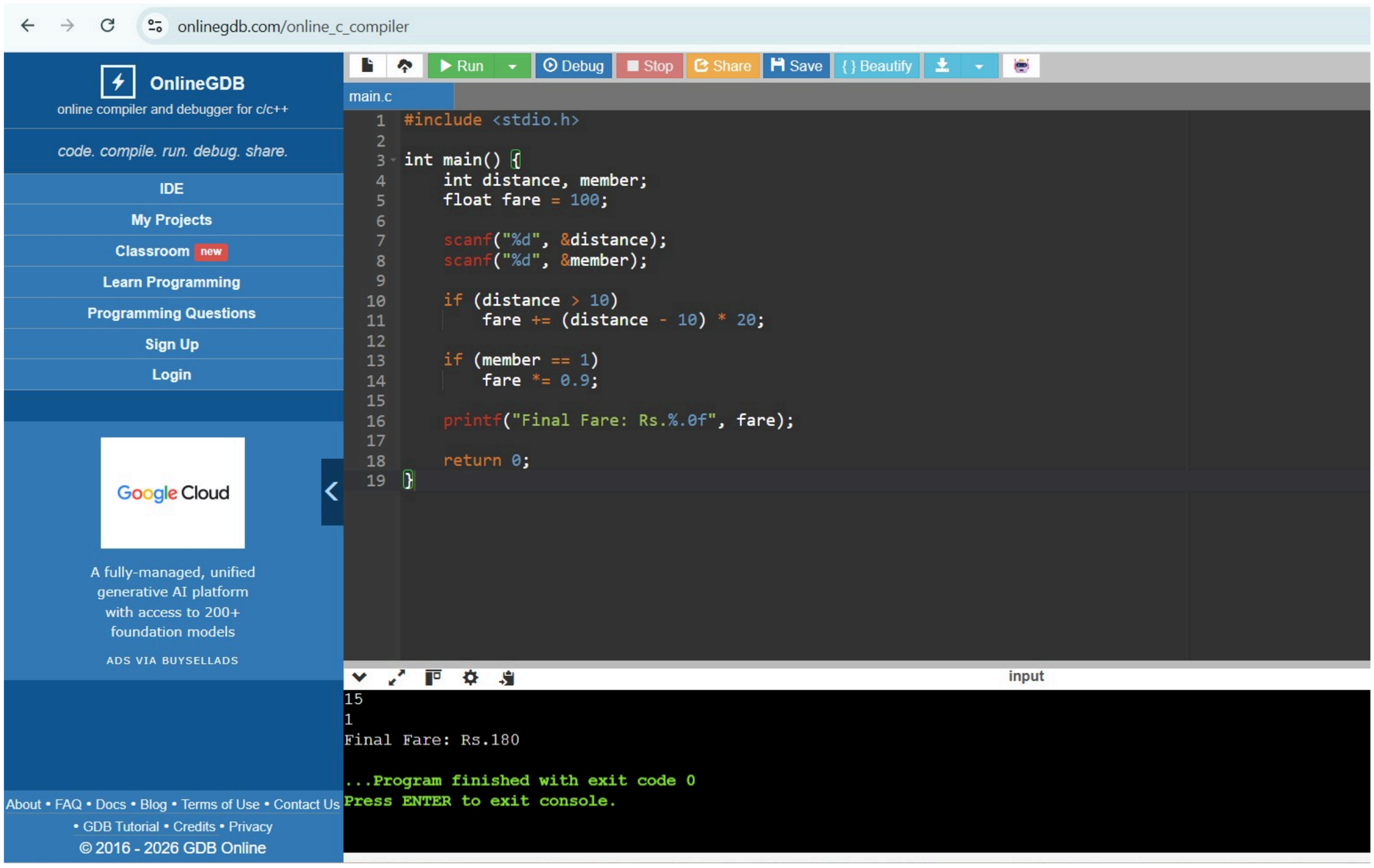
Input:

15

1

Output:

Final Fare: Rs.180



QUESTION 2:

2. Online Examination System

Aim

To check whether a student is allowed to submit an online examination.

Algorithm

1. Start.
2. Read the number of questions answered, remaining time, and submission status.

3. Check whether at least one question is answered.
4. Check whether the remaining time is greater than zero.
5. Check whether the exam has not already been submitted.
6. Display the submission status.
7. Stop.

Pseudocode

```
START

READ questions, time, submitted

IF questions >= 1 AND time > 0 AND submitted == 0

   PRINT "Submission Allowed"

ELSE

   PRINT "Submission Not Allowed"

END IF

STOP
```

C Program

```
#include <stdio.h>



int main() {

   int questions, time, submitted;



   scanf("%d", &questions);

   scanf("%d", &time);

   scanf("%d", &submitted);



if (questions >= 1 && time > 0 && submitted == 0)

   printf("Submission Allowed");
```

```
    else
        printf("Submission Not Allowed");


    return 0;
}
```

Result

Input:

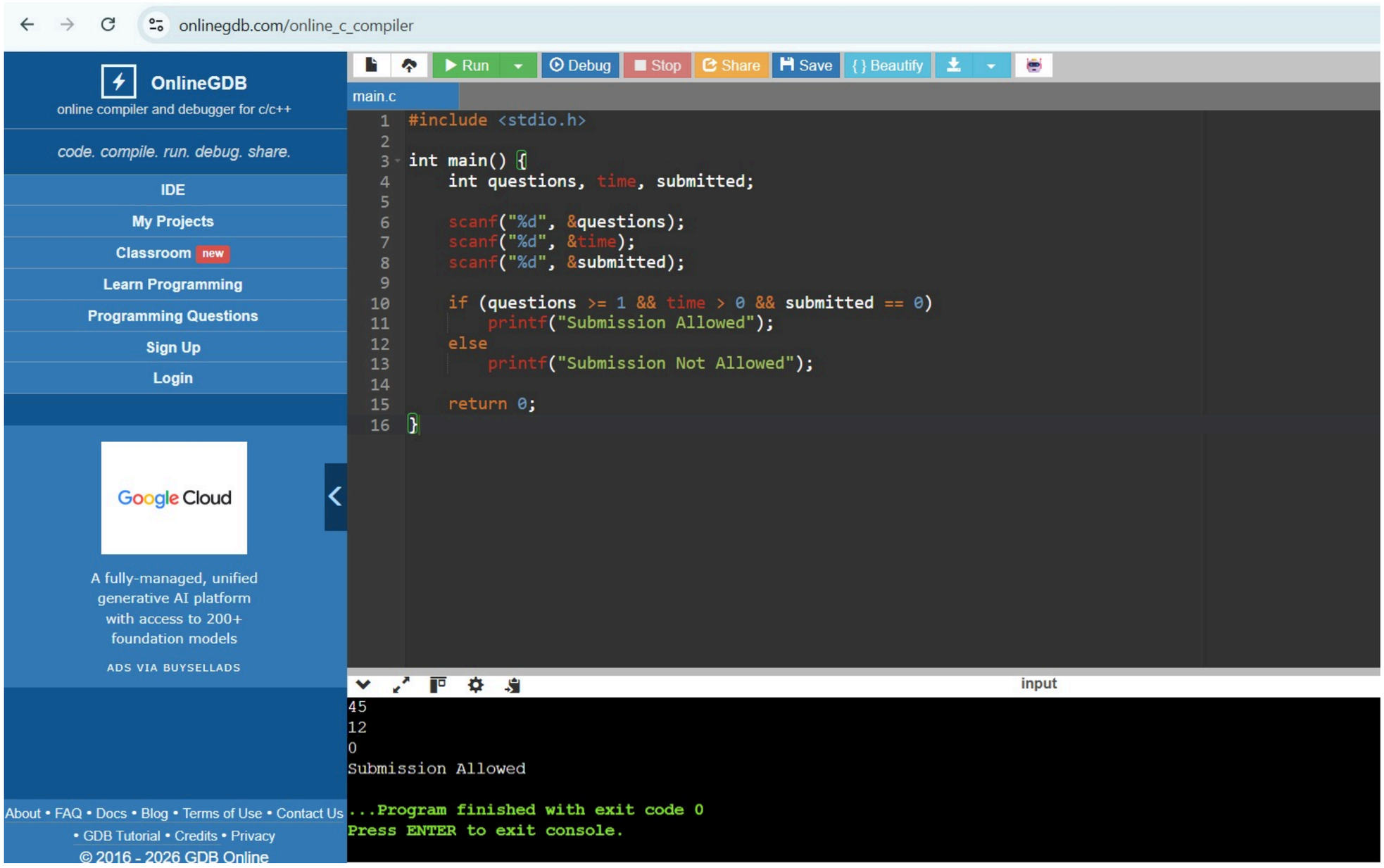
45

12

0

Output:

Submission Allowed



QUESTION 3:

3. Treasure Hunt

Aim

To calculate the shares of Blackbeard, Long Ben, and the other pirates.

Algorithm

1. Start.
2. Read the total gold coins, Long Ben's percentage, and Blackbeard's percentage.
3. Calculate Long Ben's share.
4. Subtract Long Ben's share from the total coins.
5. Calculate Blackbeard's share from the remaining coins.
6. Calculate the remaining coins for the other pirates.
7. Display the shares.
8. Stop.

Pseudocode

START

READ coins, x, y

longBen = coins * x / 100

remaining = coins - longBen

blackbeard = remaining * y / 100

others = remaining - blackbeard

PRINT longBen

PRINT blackbeard

PRINT others

STOP

C Program

```
#include <stdio.h>


int main() {
    int coins, x, y;
    int longBen, remaining, blackbeard, others;

    scanf("%d", &coins);
    scanf("%d", &x);
    scanf("%d", &y);

    longBen = coins * x / 100;
    remaining = coins - longBen;
    blackbeard = remaining * y / 100;
    others = remaining - blackbeard;

    printf("%d\n", longBen);
    printf("%d\n", blackbeard);
    printf("%d\n", others);

    return 0;
}
```

Result

Input:

729

65

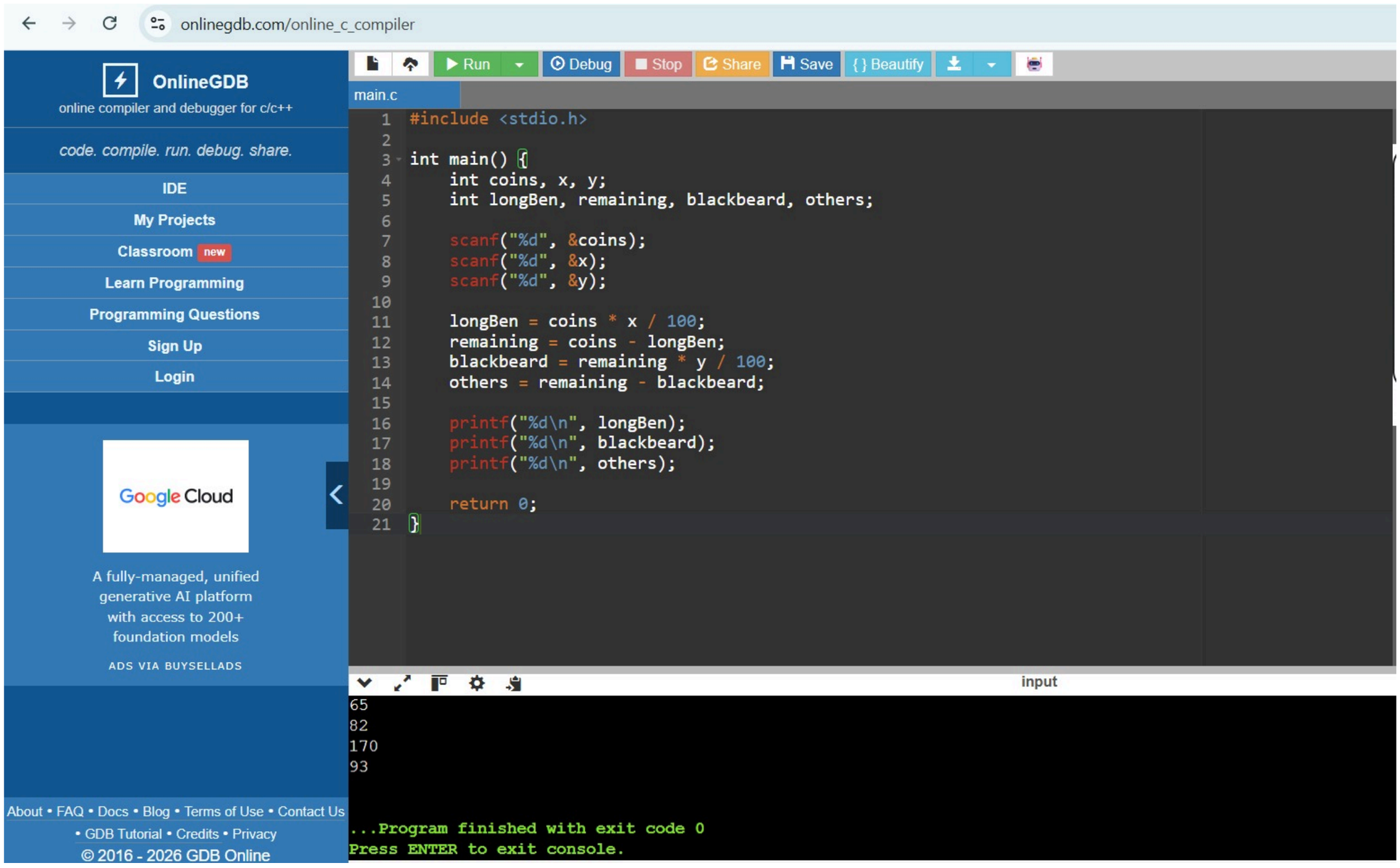
87

Output:

473

222

29



QUESTION 4:

4. Dollars and Cents

Aim

To add two amounts of dollars and cents.

Algorithm

1. Start.
2. Read the first dollar and cent values.

3. Read the second dollar and cent values.
4. Add both dollar values and both cent values.
5. Convert every 100 cents into one dollar.
6. Display the total dollars and cents.
7. Stop.

Pseudocode

```
START

READ d1, c1, d2, c2

dollars = d1 + d2

cents = c1 + c2

dollars = dollars + cents / 100

cents = cents % 100

PRINT dollars, cents

STOP
```

C Program

```
#include <stdio.h>



int main() {

    int d1, c1, d2, c2;

    int dollars, cents;

    scanf("%d%d%d%d", &d1, &c1, &d2, &c2);



    dollars = d1 + d2;

    cents = c1 + c2;
```

```
dollars += cents / 100;

cents %= 100;


printf("%d\n%d", dollars, cents);

    return 0;

}
```

Result

Input:

30

10

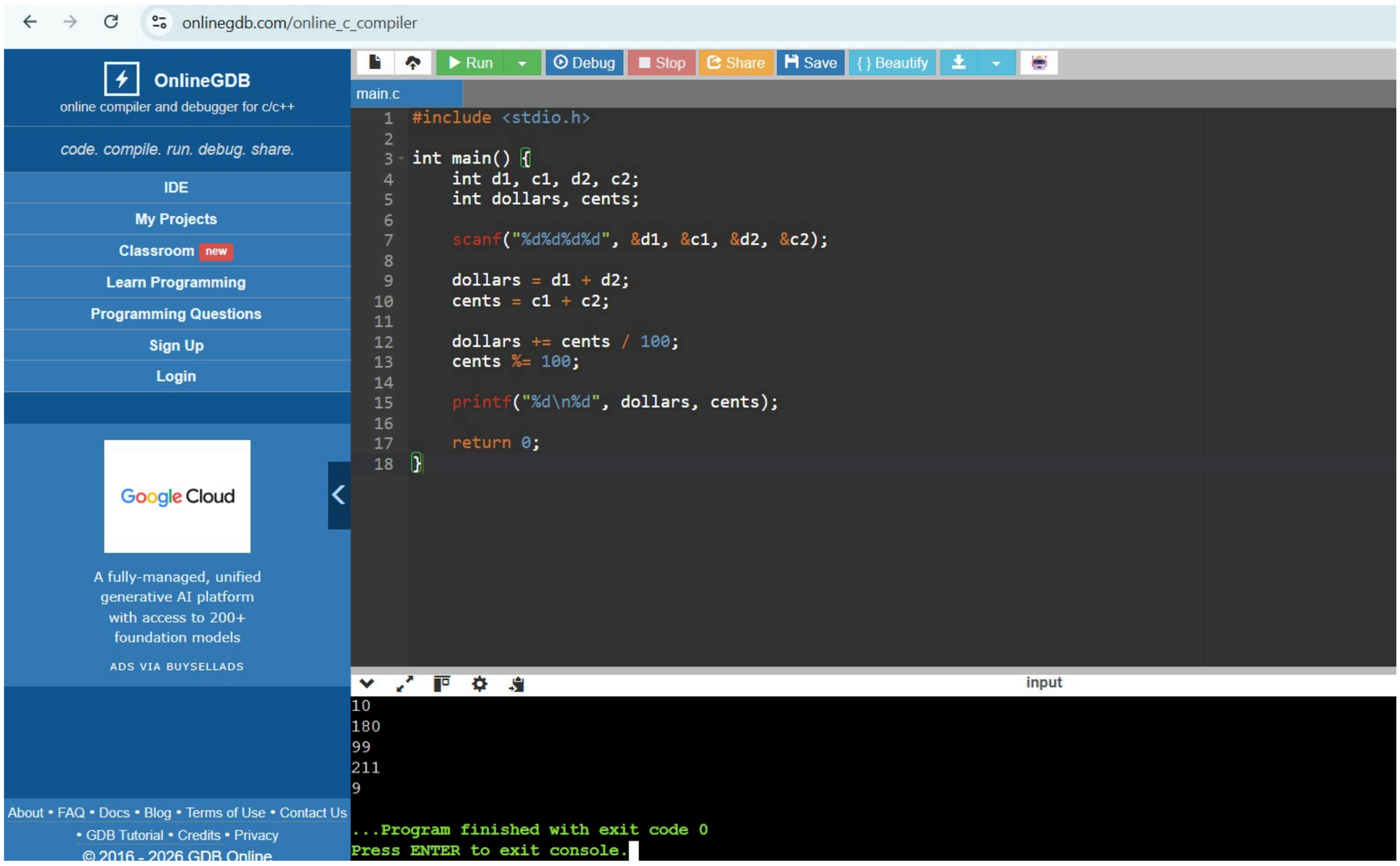
140

99

Output:

171

9



QUESTION 5:

5. Car Mileage

Aim

To determine whether a car can reach its destination using the available petrol.

Algorithm

1. Start.
2. Read mileage, petrol available, and distance to travel.
3. Calculate the possible distance using mileage multiplied by petrol.
4. If the possible distance is greater than or equal to the required distance, display "Can reach".
5. Otherwise, display "Cannot reach".
6. Stop.

Pseudocode

```
START

READ mileage, petrol, distance

possibleDistance = mileage * petrol

IF possibleDistance >= distance
   PRINT "Can reach"
ELSE
   PRINT "Cannot reach"
END IF

STOP
```

C Program

```
#include <stdio.h>

int main() {
    float mileage;
    int petrol, distance;

    scanf("%f%d%d", &mileage, &petrol, &distance);

    if (mileage * petrol >= distance)
        printf("Can reach");
    else
        printf("Cannot reach");

    return 0;
}
```

Result

Input:

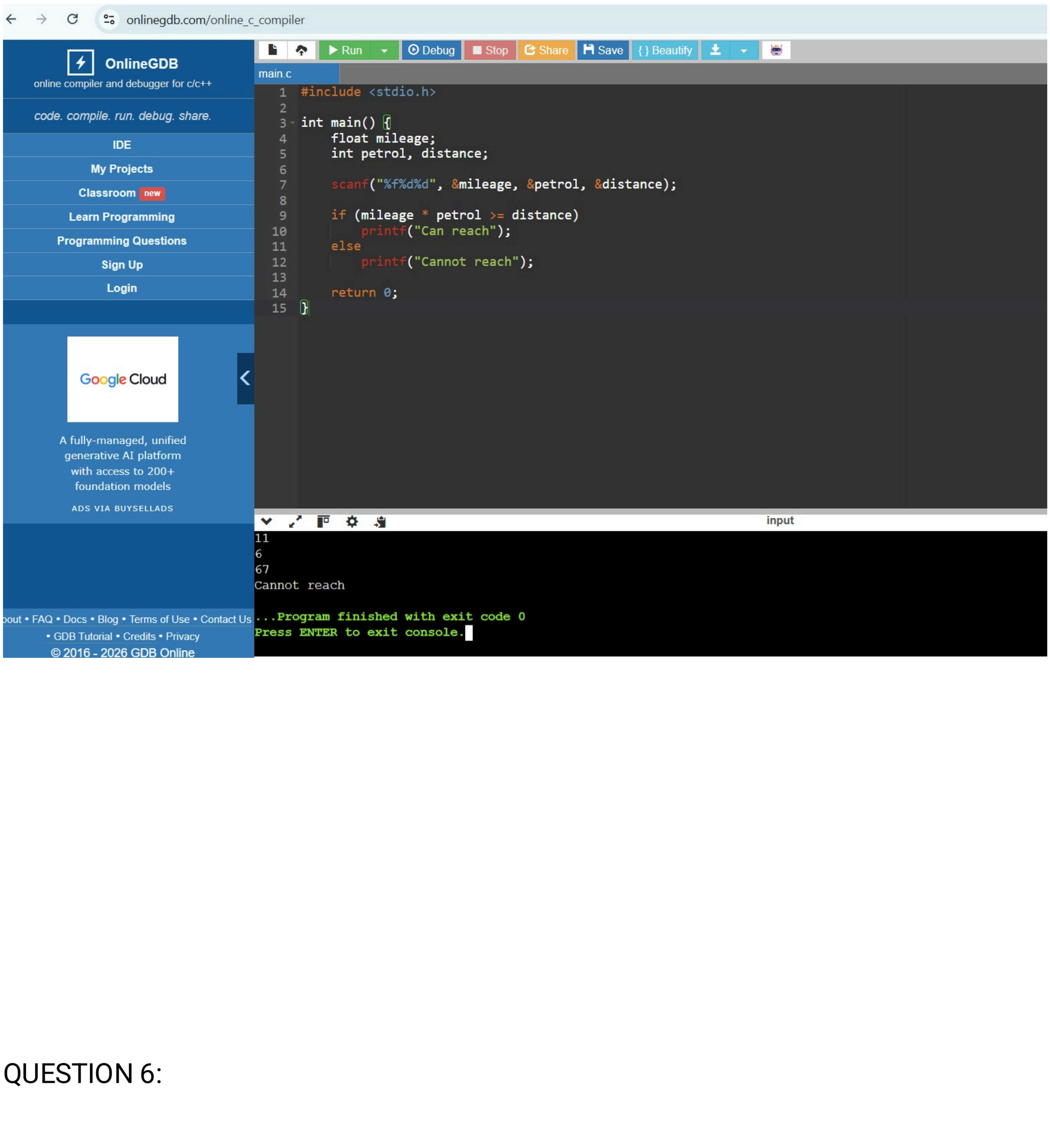
11

6

67

Output:

Cannot reach



QUESTION 6:

6. Car Race

Aim

To determine which road the car will travel on based on the given road numbers and the number of cars.

Algorithm

1. Start.
2. Read the numbers of roads A, B, and C.
3. Read the number of cars.
4. Check the car's position using the given road numbers.
5. Display the road selected by the car.
6. If no road is available, display "No path exists".
7. Stop.

Pseudocode

```
START

READ A, B, C, cars

IF cars MOD 3 == 1

   PRINT "Car 1 goes into road A"

ELSE IF cars MOD 3 == 2

   PRINT "Car 1 goes into road B"

ELSE IF cars MOD 3 == 0

   PRINT "Car 1 goes into road C"

ELSE

   PRINT "No path exists"

END IF

STOP
```

C Program

```
#include <stdio.h>


int main() {

    int A, B, C, cars;

scanf("%d%d%d%d", &A, &B, &C, &cars);


if (cars % 3 == 1)

    printf("Car 1 goes into road A");

else if (cars % 3 == 2)

    printf("Car 1 goes into road B");

else if (cars % 3 == 0)

        printf("Car 1 goes into road C");

    else

        printf("No path exists");


    return 0;

}
```

Result

Input:

243

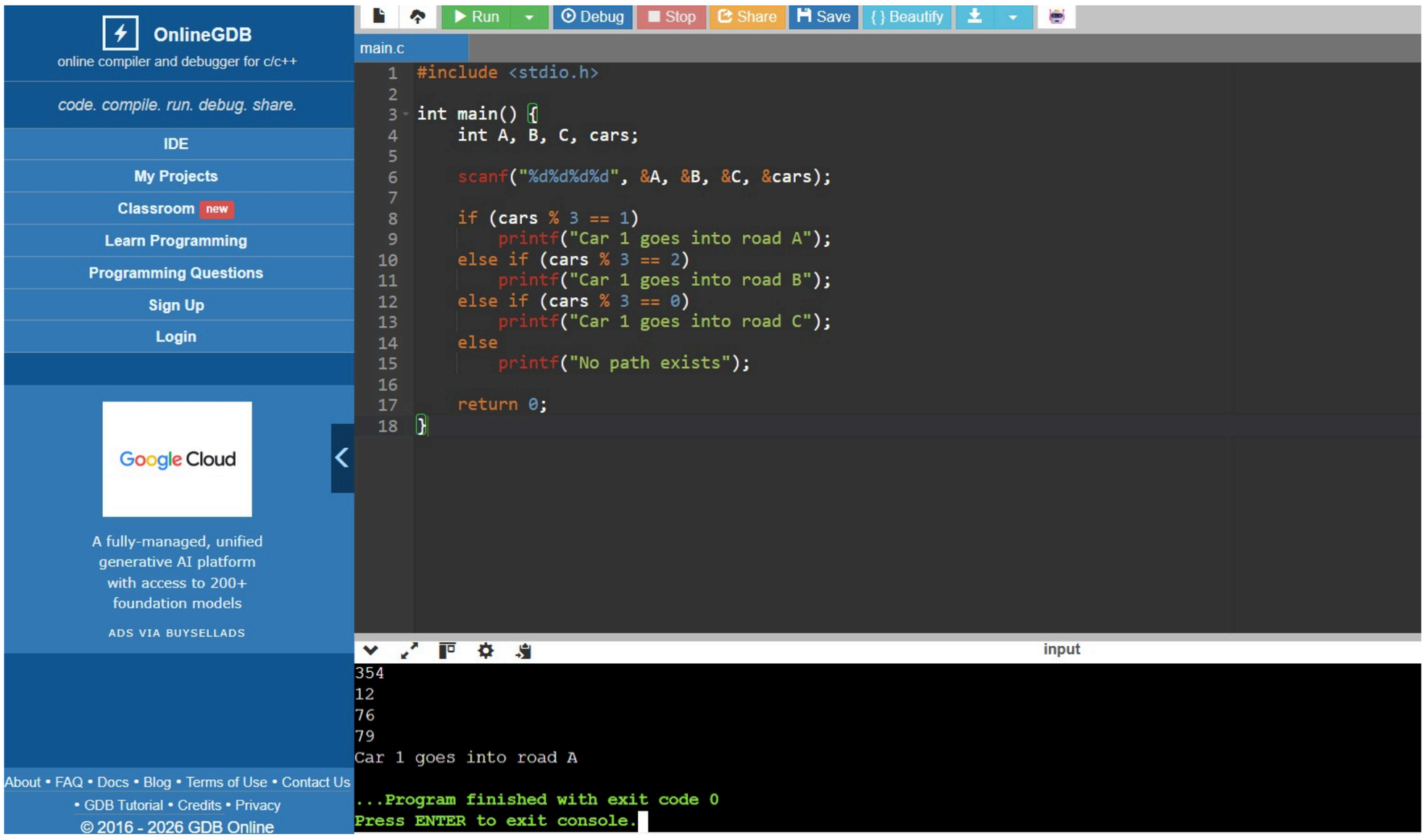
136

567

17

Output:

Car 1 goes into road B



QUESTION 7:

7. Debt Repay

Aim

To calculate the interest, total amount, discount, and final settlement amount.

Algorithm

1. Start.
2. Read the principal amount, interest rate, and number of years.
3. Calculate simple interest.
4. Calculate the total amount by adding interest to the principal.
5. Calculate a 2% discount on the interest.
6. Subtract the discount from the total amount.
7. Display all values to two decimal places.

8. Stop.

Pseudocode

```
START

READ principal, rate, years

interest = principal * rate * years / 100

amount = principal + interest

discount = interest * 0.02

finalAmount = amount - discount

PRINT interest

PRINT amount

PRINT discount

PRINT finalAmount

STOP
```

C Program

```
#include <stdio.h>



int main() {

    float principal, rate, years;

    float interest, amount, discount, finalAmount;



    scanf("%f%f%f", &principal, &rate, &years);

interest = principal * rate * years / 100;

amount = principal + interest;

discount = interest * 0.02;

finalAmount = amount - discount;
```

```
    printf("%.2f\n", interest);

    printf("%.2f\n", amount);

    printf("%.2f\n", discount);

    printf("%.2f\n", finalAmount);


    return 0;
}
```

Result

Input:

```
100

1

10
```

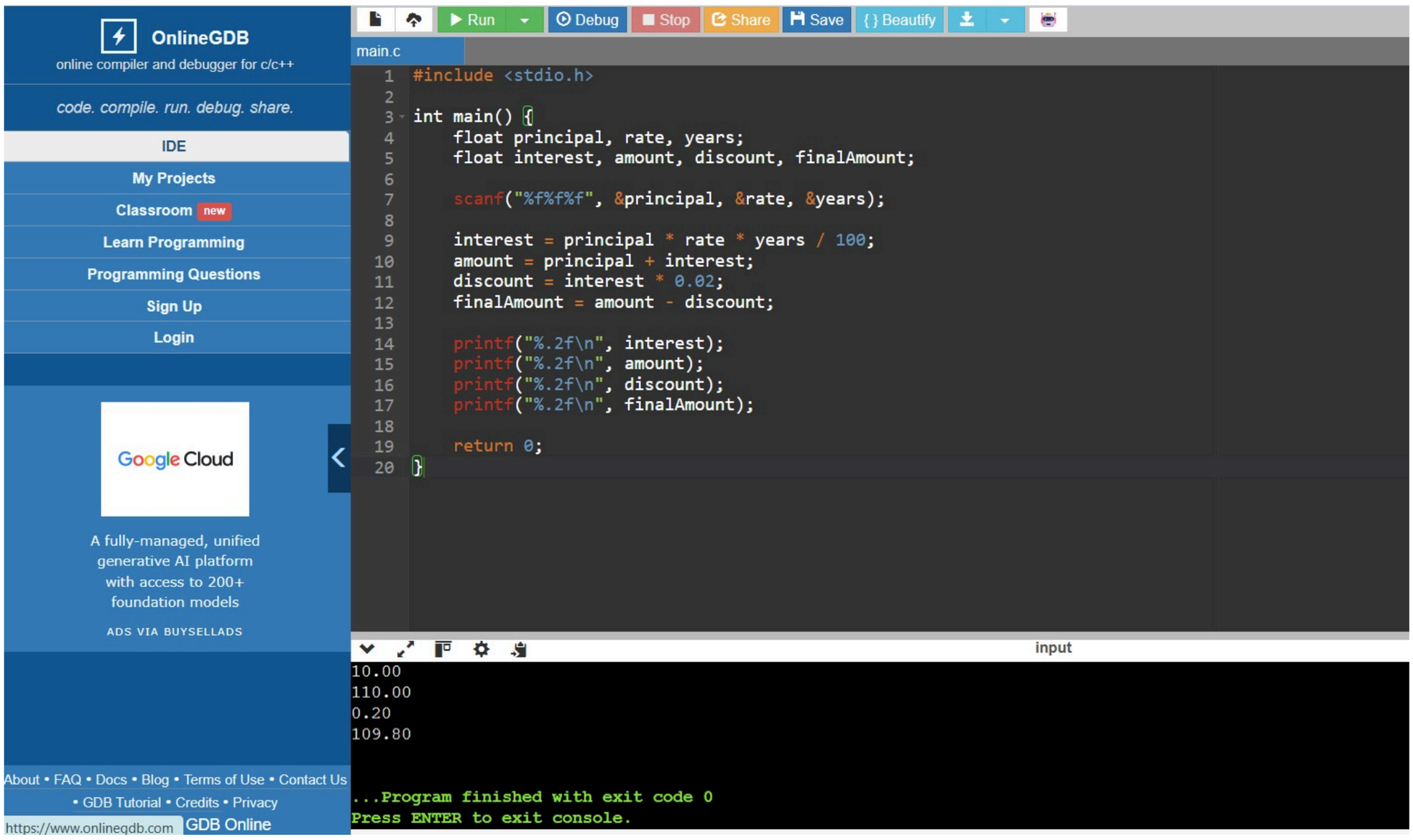
Output:

```
10.00

110.00

0.20

109.80
```



QUESTION 8:

8. Employee Job Level

Aim

To determine an employee's job level based on experience and performance score.

Algorithm

1. Start.
2. Read the employee's years of experience and performance score.
3. If experience is greater than 7 years and performance is at least 85, classify as Senior.

4. Otherwise, if experience is between 3 and 7 years and performance is at least 80, classify as Mid-Level.
5. Otherwise, classify as Junior.
6. Display the job level.
7. Stop.

Pseudocode

```
START
READ experience, performance
IF experience > 7 AND performance >= 85
   PRINT "Level: Senior"
ELSE IF experience >= 3 AND experience <= 7 AND performance >= 80
   PRINT "Level: Mid-Level"
ELSE
   PRINT "Level: Junior"
END IF
STOP
```

C Program

```
#include <stdio.h>


int main() {
   int experience, performance;

scanf("%d%d", &experience, &performance);


if (experience > 7 && performance >= 85)
   printf("Level: Senior");
```

```
else if (experience >= 3 && experience <= 7 && performance >= 80)
    printf("Level: Mid-Level");
else
    printf("Level: Junior");

    return 0;
}
```

Result

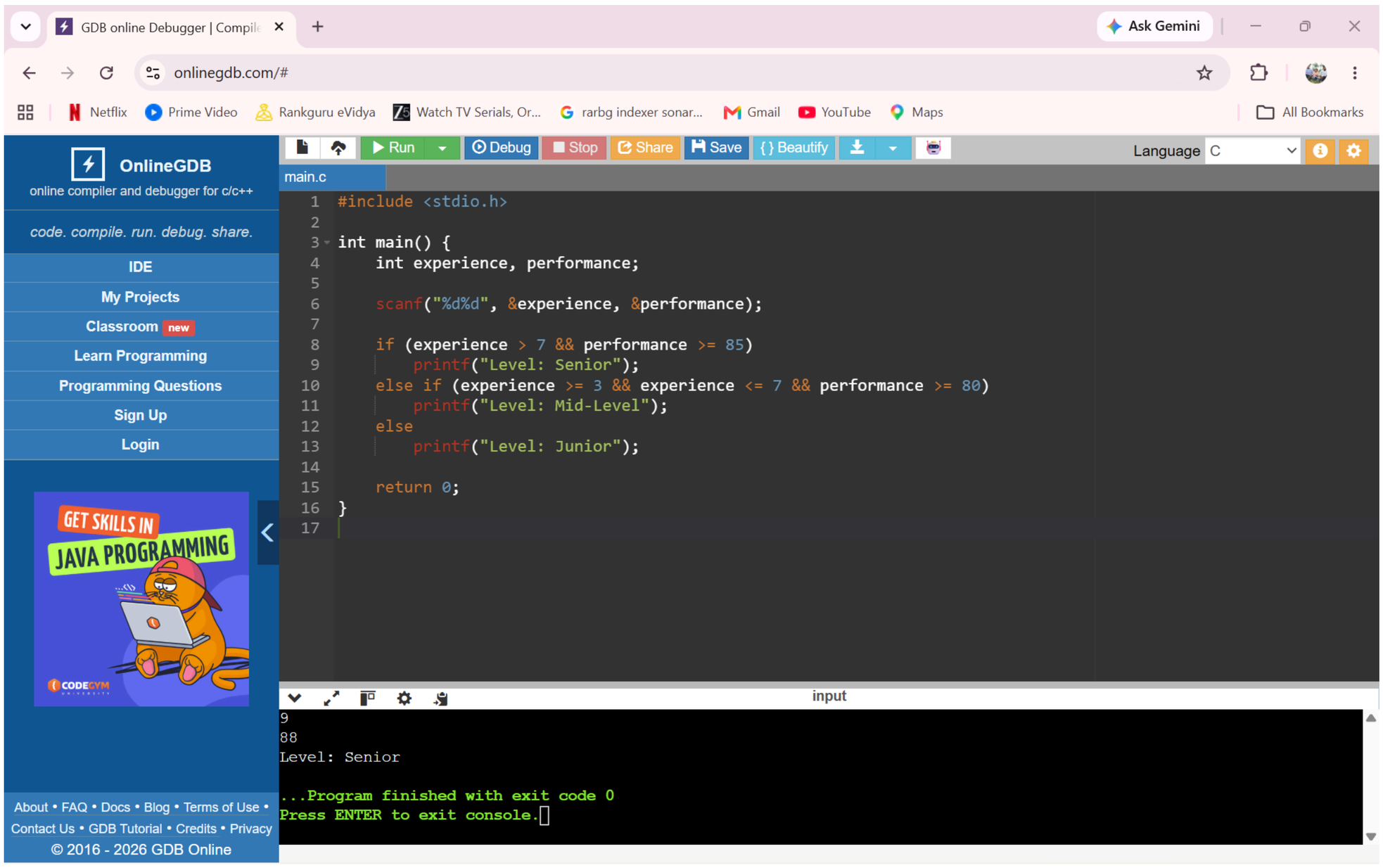
Input:

9

88

Output:

Level: Senior



QUESTION 9:

9. Number of Days

Aim

To find the number of days in a given month and year.

Algorithm

1. Start.
2. Read the year and month.
3. If the year is outside the range 1900– 9999, display 0.
4. Otherwise, check the month.
5. For February, check whether the year is a leap year.
6. Assign the appropriate number of days.
7. Display the result.
8. Stop.

Pseudocode

```
START

READ year, month

IF year < 1900 OR year > 9999

   PRINT 0

ELSE

   IF month == 2

IF year % 400 == 0 OR

  (year % 4 == 0 AND year % 100 != 0)

   days = 29

ELSE

   days = 28
```

```
    END IF
ELSE IF month == 4 OR 6 OR 9 OR 11
    days = 30
ELSE IF month >= 1 AND month <= 12
    days = 31
    ELSE
      days = 0
    END IF
    PRINT days, "Days"
END IF
STOP
```

C Program

```
#include <stdio.h>

int main() {
    int year, month, days;

    scanf("%d%d", &year, &month);

    if (year < 1900 || year > 9999) {
        printf("0");
        return 0;
    }

    if (month == 2) {
        if (year % 400 == 0 ||
```

```
        (year % 4 == 0 && year % 100 != 0))
        days = 29;
    else
        days = 28;
}
else if (month == 4 || month == 6 ||
        month == 9 || month == 11)
    days = 30;
else if (month >= 1 && month <= 12)
    days = 31;
else
    days = 0;

    if (days == 0)
        printf("0");
    else
        printf("%d Days", days);


    return 0;
}
```

Result

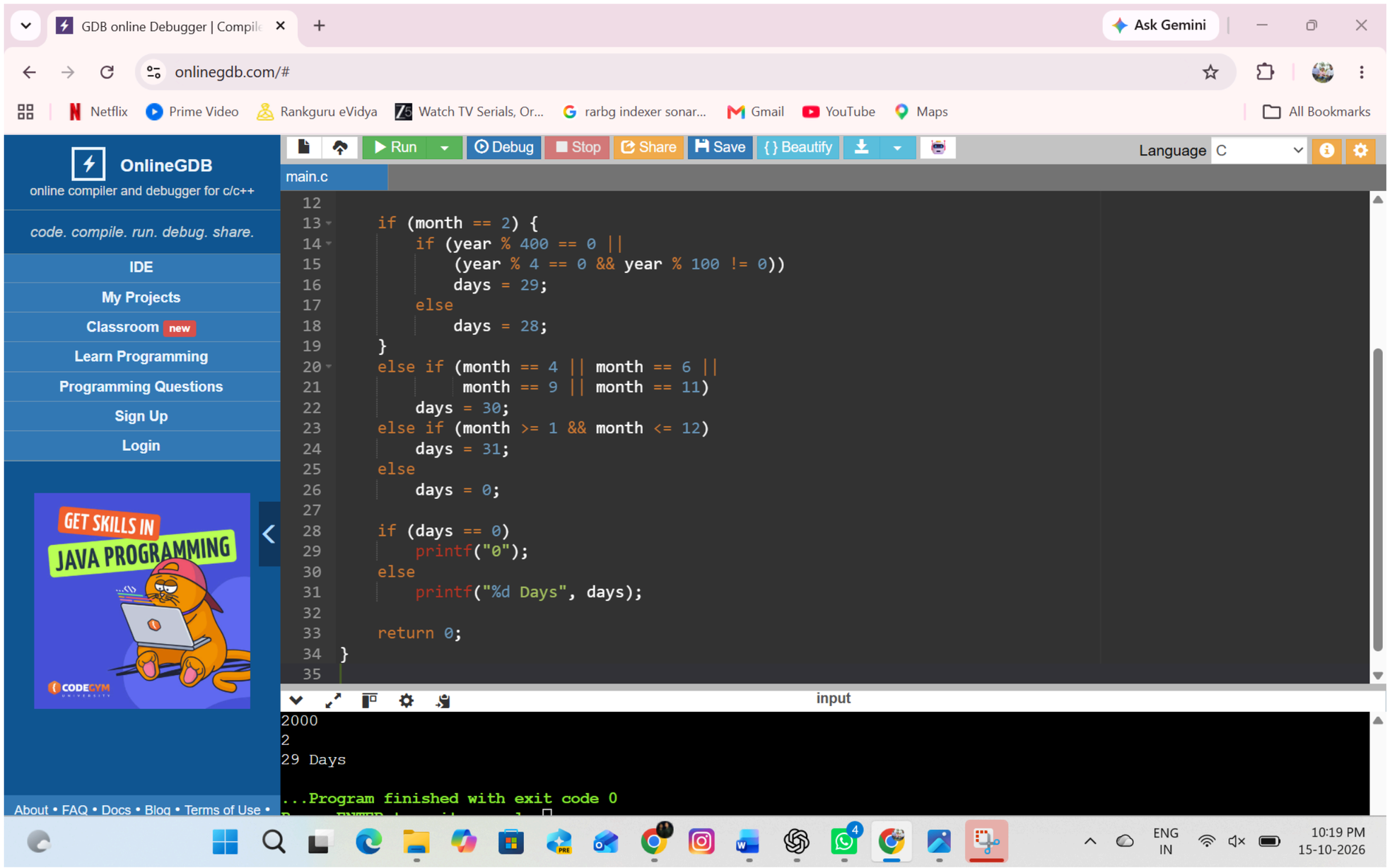
Input:

2000

2

Output:

29 Days



QUESTION 10:

10. Electricity Bill

Aim

To calculate the electricity bill based on the number of units consumed.

Algorithm

1. Start.
2. Read the units consumed.
3. If units are less than or equal to 200, calculate the bill at ₹ 0.50 per unit.
4. If units are less than or equal to 400, calculate the bill at ₹ 0.65 per unit and add ₹ 100.
5. If units are less than or equal to 600, calculate the bill at ₹ 0.80 per unit and add ₹ 200.
6. Otherwise, calculate the bill at ₹ 1.25 per unit and add ₹ 425.

7. Display the electricity bill.
8. Stop.

Pseudocode

```
START

READ units

IF units <= 200

   bill = units * 0.50

ELSE IF units <= 400

   bill = units * 0.65 + 100

ELSE IF units <= 600

   bill = units * 0.80 + 200

ELSE

   bill = units * 1.25 + 425

END IF

PRINT bill

STOP
```

C Program

```
#include <stdio.h>


int main() {

   int units;

   float bill;


   scanf("%d", &units);


   if (units <= 200)
```

```
    bill = units * 0.50;
else if (units <= 400)
    bill = units * 0.65 + 100;
else if (units <= 600)
    bill = units * 0.80 + 200;
    else
        bill = units * 1.25 + 425;


    printf("Rs.%.2f", bill);


    return 0;
}
```

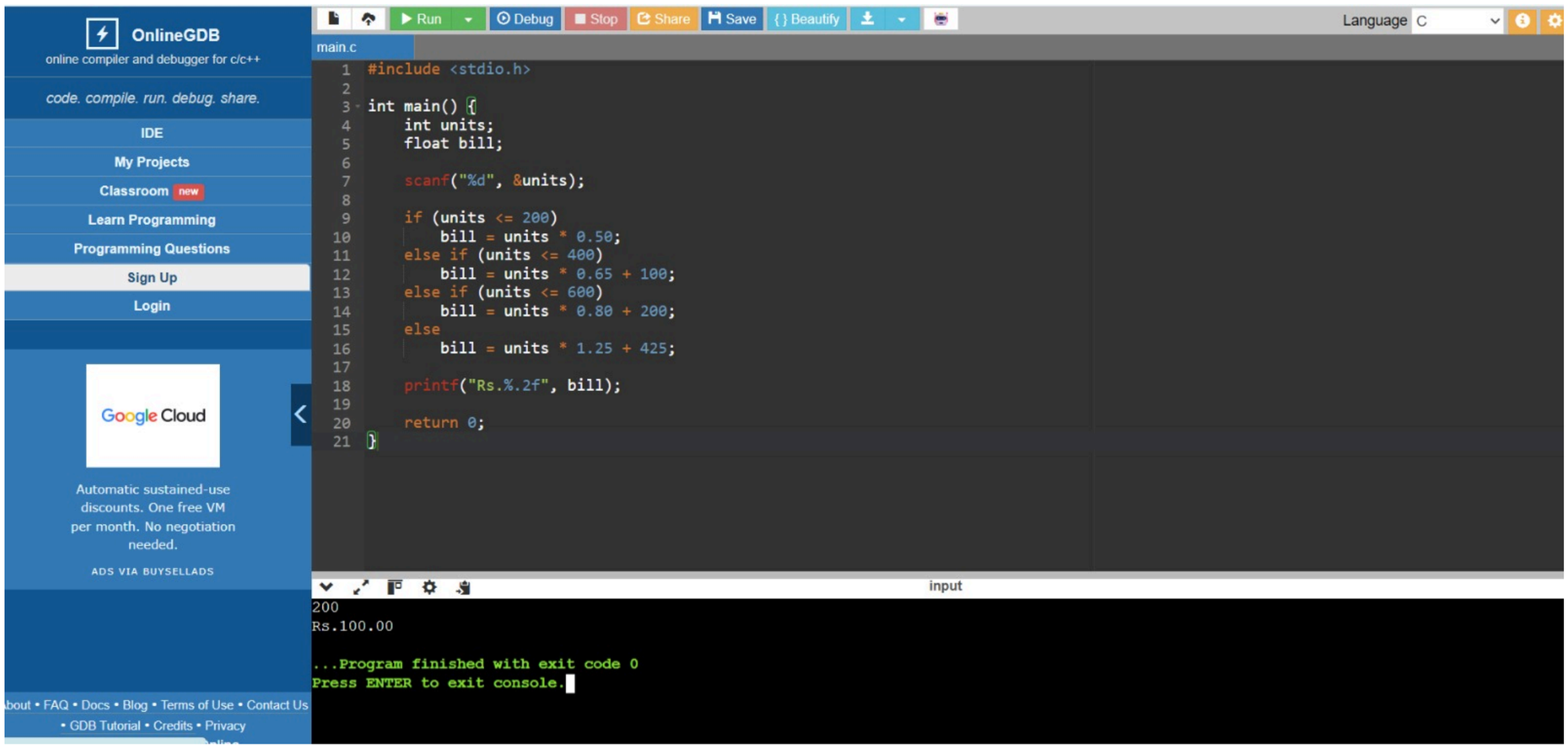
Result

Input:

200

Output:

Rs.100.00


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Debug
Stop
Share
Save
{ } Beautify
Language C
main.c
#include <stdio.h>
int main() {
int units;
float bill;
scanf("%d", &units);
if (units <= 200)
bill = units * 0.50;
else if (units <= 400)
bill = units * 0.65 + 100;
else if (units <= 600)
bill = units * 0.80 + 200;
else
bill = units * 1.25 + 425;
printf("Rs.%.2f", bill);
return 0;
}
input
200
Rs.100.00
...Program finished with exit code 0
Press ENTER to exit console.